

End-to-End Automated Neonatal EEG Seizure Detection Pipeline



Core Innovation: Ultra-Compact Biomarker

- 12 parameters per channel vs. 432 for EMD / 24 for DWT
- Captures physical 1/f spectral roll-off tilt and amplitude burst
- Closed-form analytical OLS solution requires < 0.5 ms per epoch

Rigorous Evaluation & Real-Time Alerting

- Strict 10-trial patient-level cross-validation (no data leakage)
- Peak Test AUROC = 0.858, Test F1 = 0.731 (outperforming EMD)
- Causal sliding filter ($W = 1 \dots 20$ s) eliminates false-alarm spikes